

A PbWO₄ SCINTILLATOR RANGE-TELESCOPE FOR DEPTH-DOSE MEASUREMENTS IN LOW-FLUX ENVIRONMENTS

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Faculty Disclosure

X	No, nothing to disclose
	Yes, please specify:

Company Name	Honoraria/ Expenses	Consulting/ Advisory Board	Funded Research	Royalties/ Patent	Stock Options	Ownership/ Equity Position	Employee	Other (please specify)

Why Low-Dose Environments?

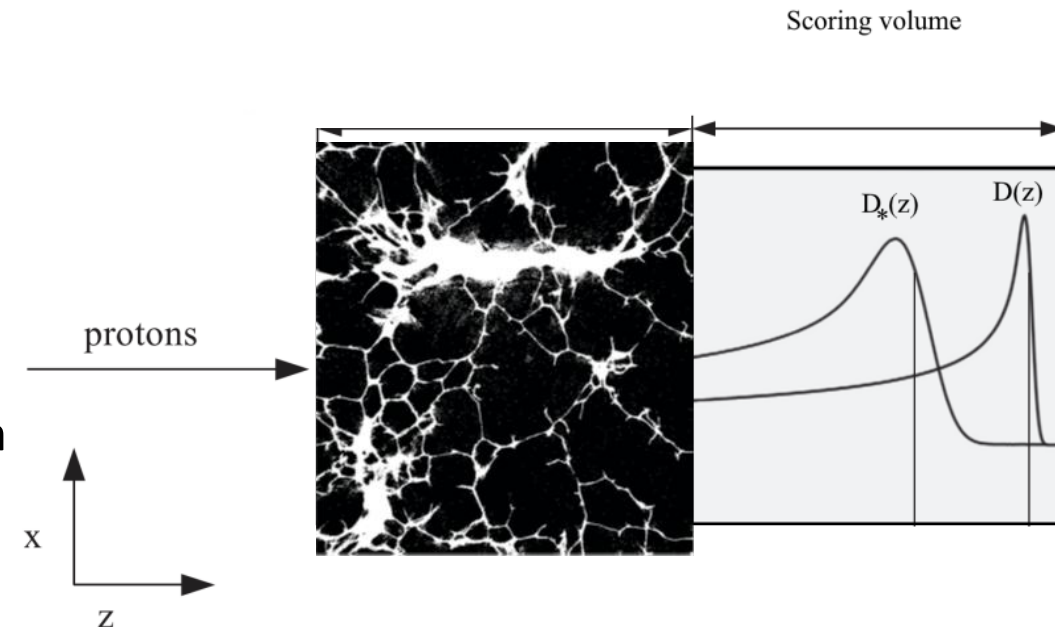
- Conventional dosimetry systems are designed for high-dose environments
 - Water columns (PTWPeakFinder)
 - MLIC (IBA Dosimetry Giraffe & Zebra)

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- Low-flux environments offer individual particle tracking
 - Useful for detector development, Monte Carlo validation

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- Low-flux environments offer individual particle tracking
 - Useful for detector development, Monte Carlo validation
 - Direct low-dose measurements of heterogeneous phantoms



Detector Design

- Segmented scintillation detector
 - Measure individual particle energy depositions as a function of depth

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- Scintillator PbWO_4
- Readout using silicon photomultipliers (SiPMs) with custom PCB and digitizers



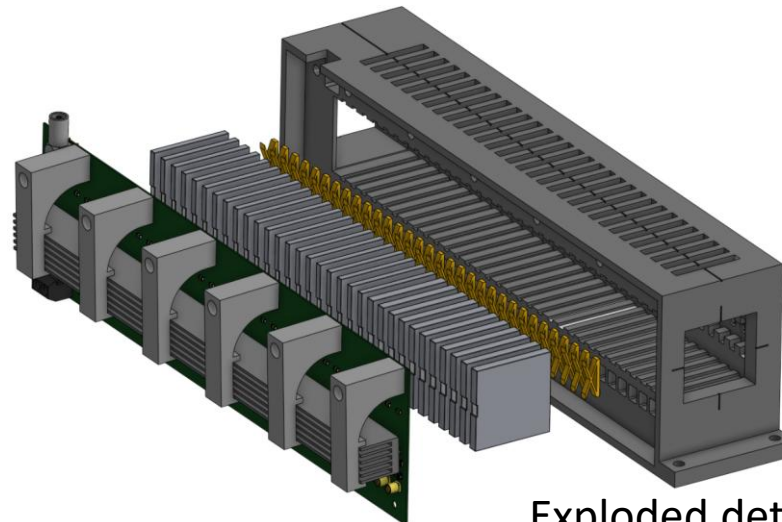
Wrapped PbWO_4 -Crystal

Detector Design

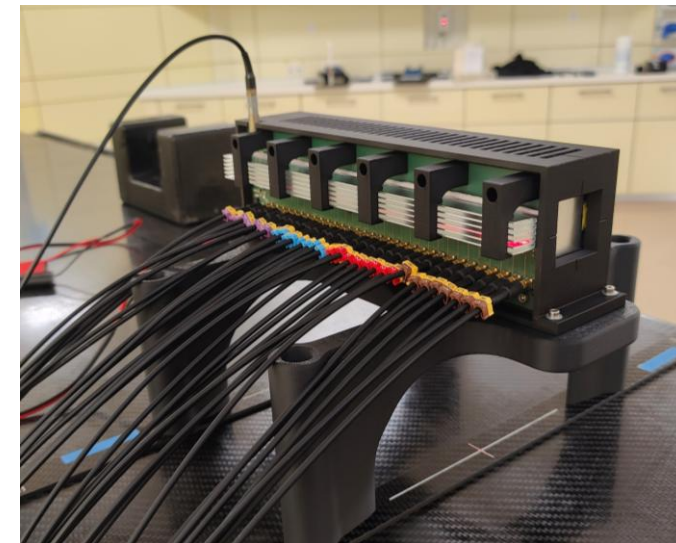
- Segmented scintillation detector
 - Measure individual particle energy depositions as a function of depth
- Scintillator PbWO_4
- Readout using silicon photomultipliers (SiPMs) with custom PCB and digitizers
- Optical coupling with Baysilone Fluid M and pressure using 3D-printed springs



Wrapped PbWO_4 -Crystal

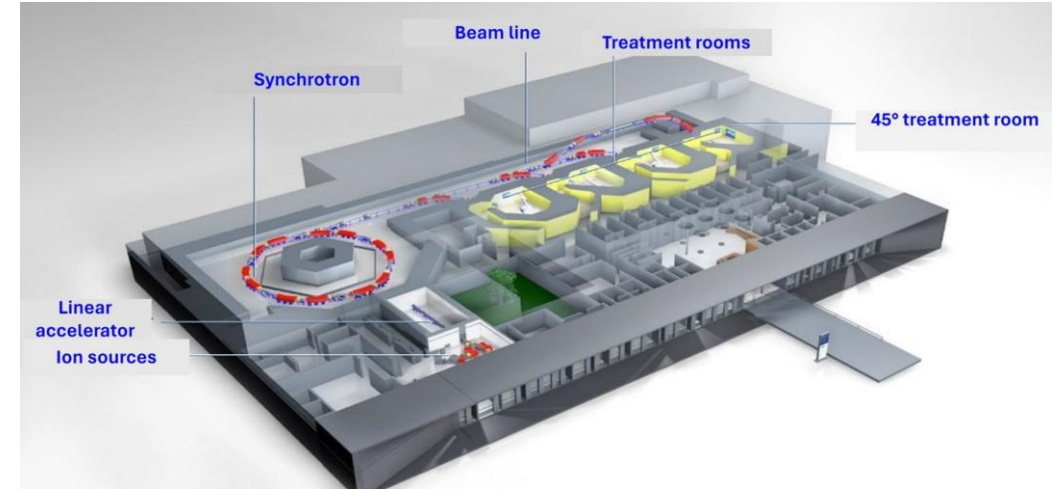


Exploded detector
Schematic & Prototype



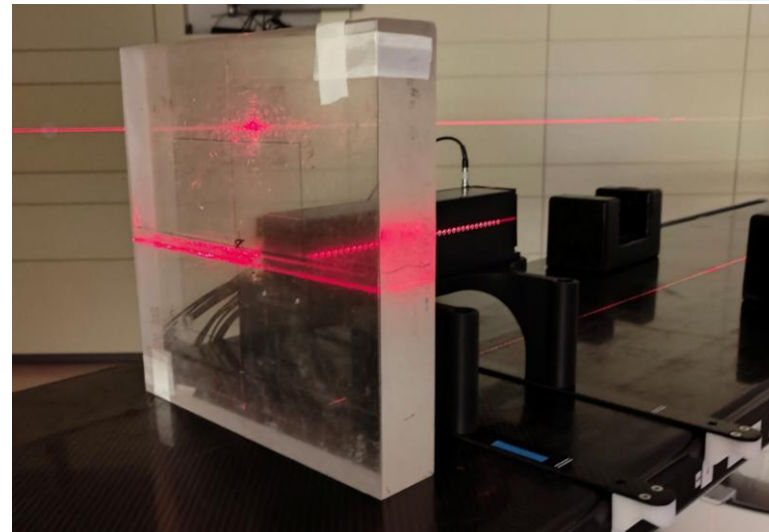
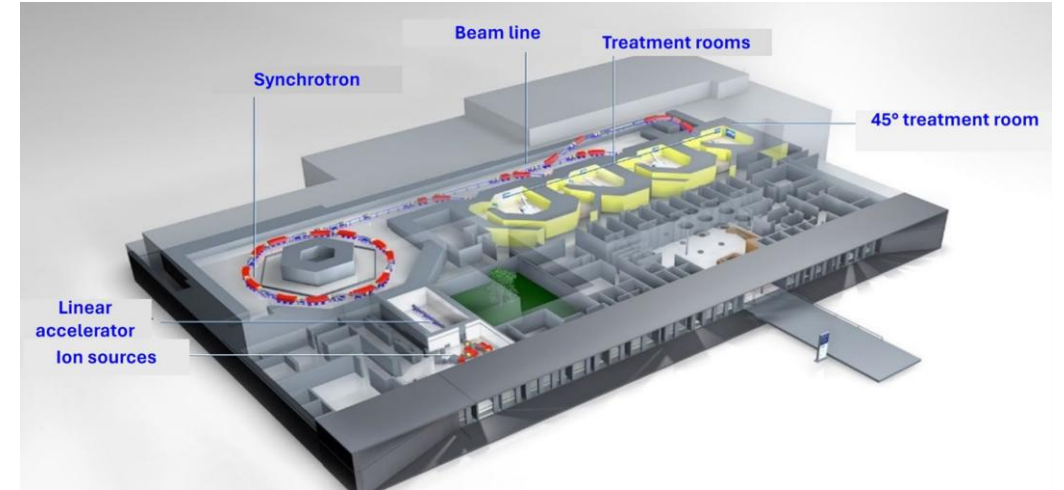
Beam Tests

- Measurements at Ion-Beam Therapy Center (MIT) in Marburg



Beam Tests

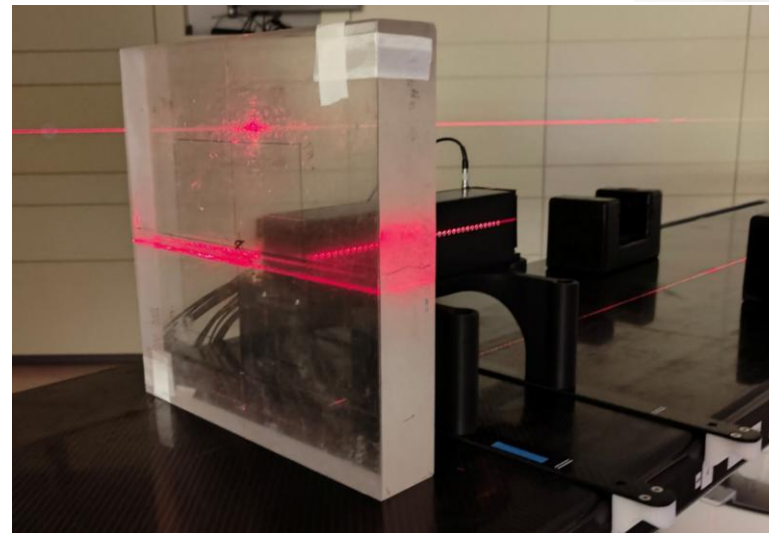
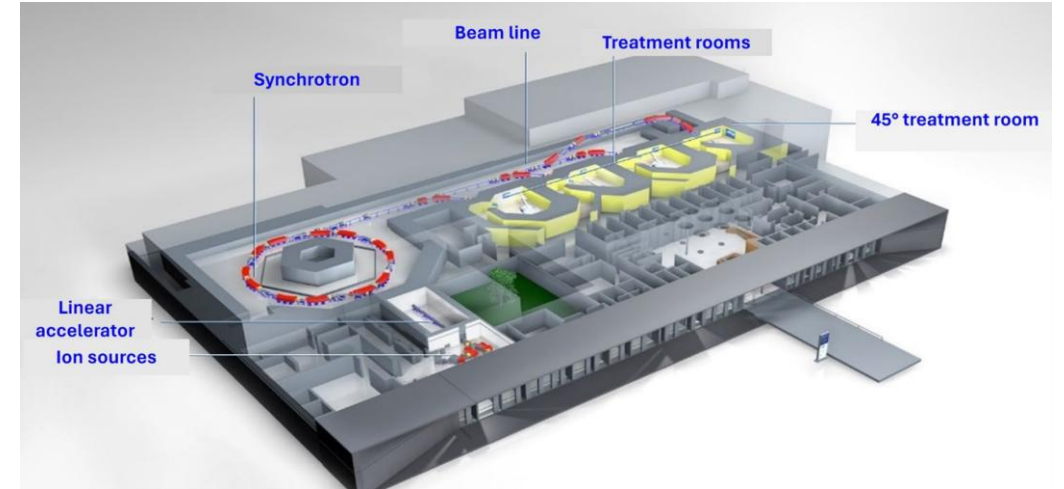
- Measurements at Ion-Beam Therapy Center (MIT) in Marburg
 - Homogeneous PMMA phantom 52 mm to 57 mm in 1 mm steps



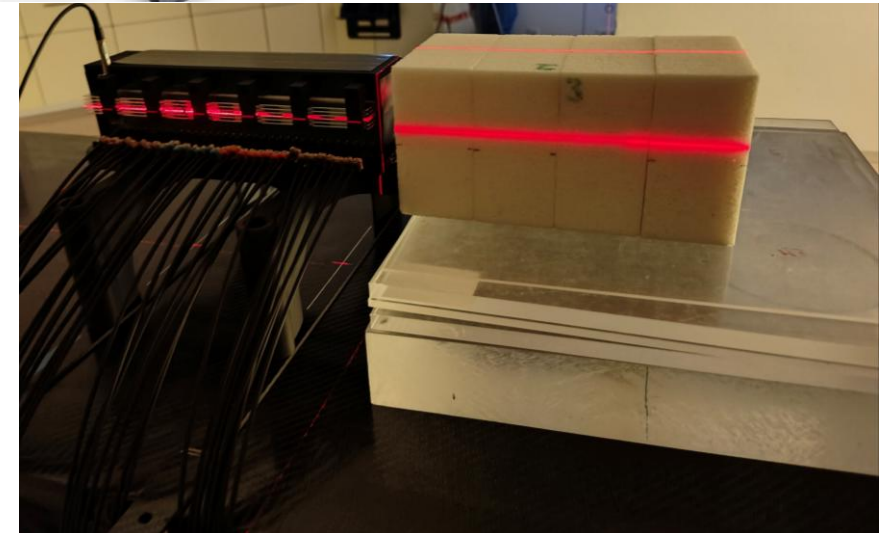
PMMA Phantom

Beam Tests

- Measurements at Ion-Beam Therapy Center (MIT) in Marburg
 - Homogeneous PMMA phantom 52 mm to 57 mm in 1 mm steps
 - Heterogeneous LN300 phantom 200 mm



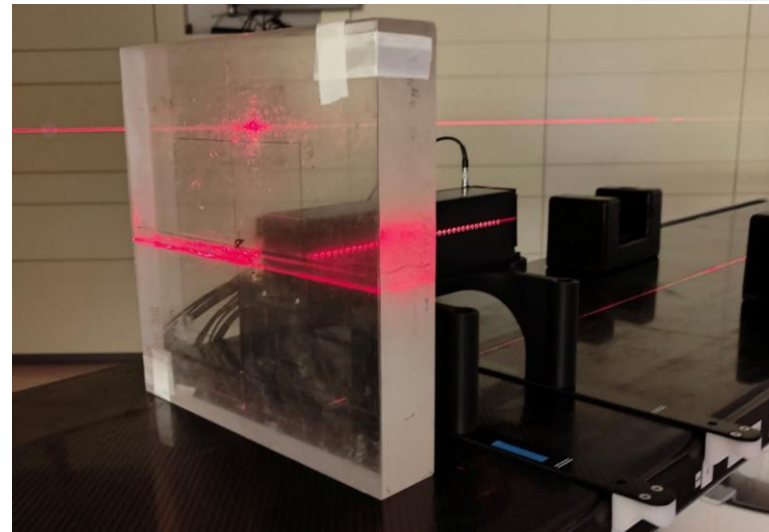
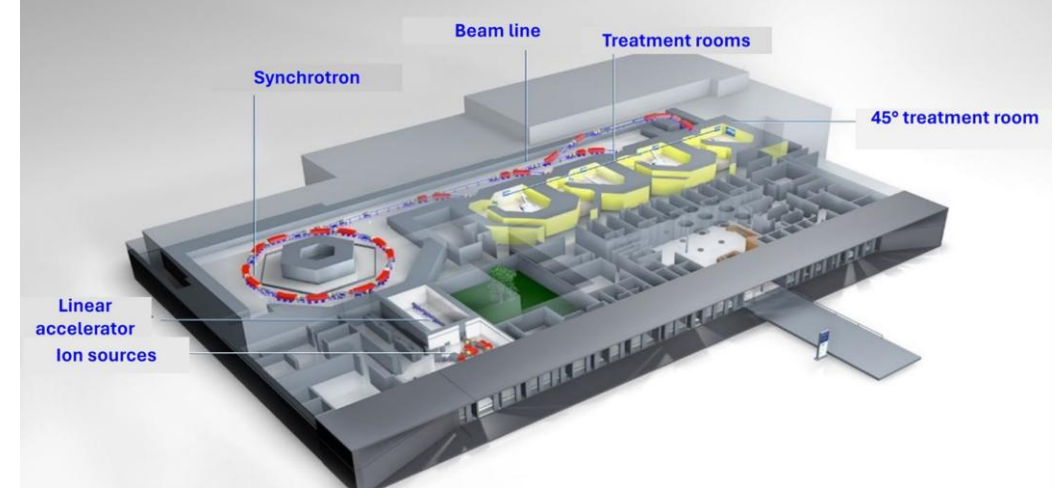
PMMA Phantom



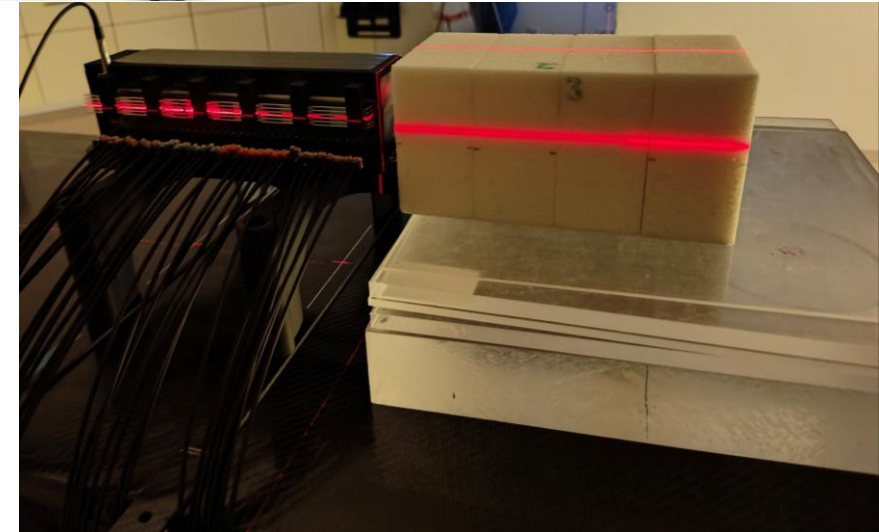
LN300 Phantom

Beam Tests

- Measurements at Ion-Beam Therapy Center (MIT) in Marburg
 - Homogeneous PMMA phantom 52 mm to 57 mm in 1 mm steps
 - Heterogeneous LN300 phantom 200 mm
- Proton beam at 221.6 MeV with particle flux of $2300 \frac{1}{s}$



PMMA Phantom



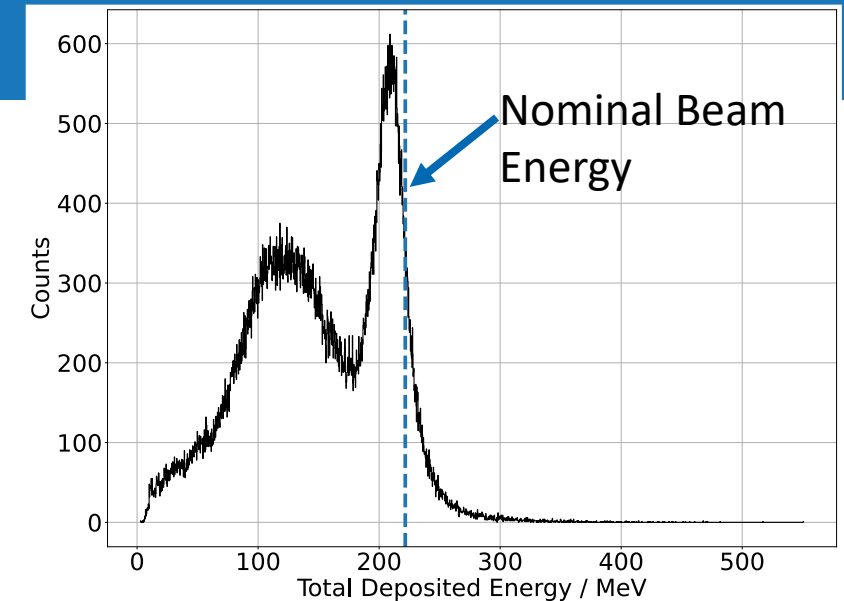
LN300 Phantom

Data Analysis

- Protons classified as multiple energy deposition events within a 50 ns global coincidence window
 - Main peak of total energy deposition using in analysis

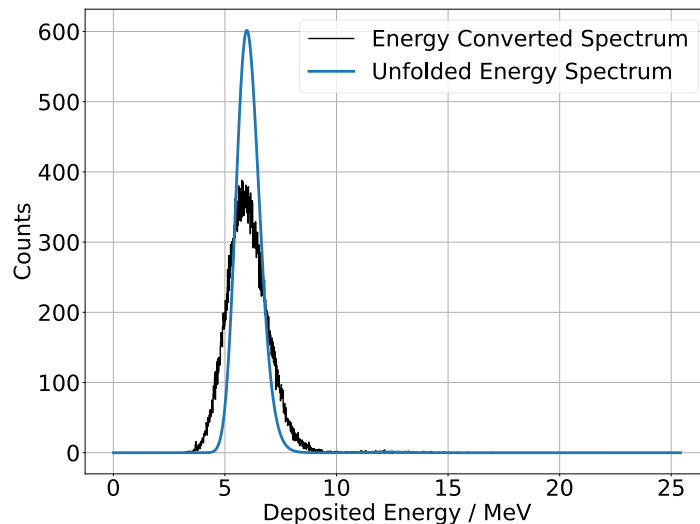
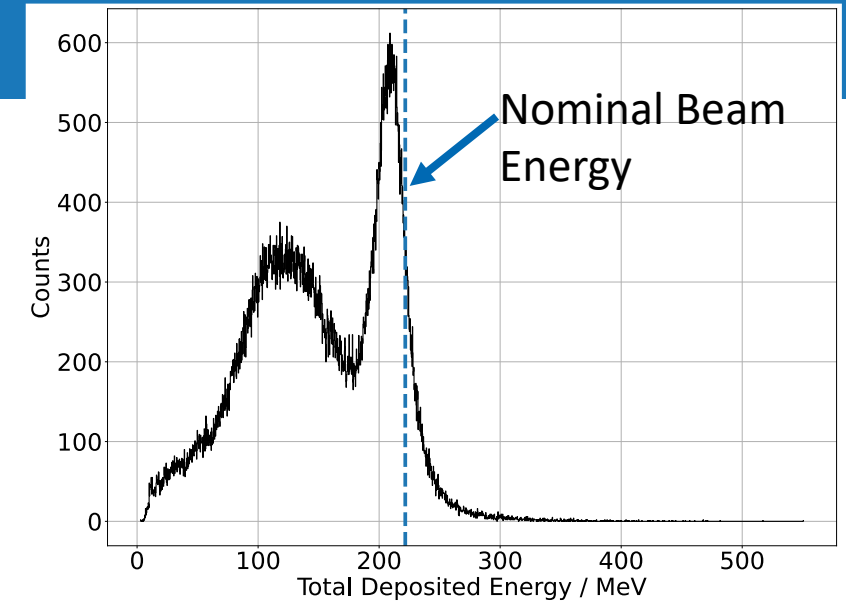
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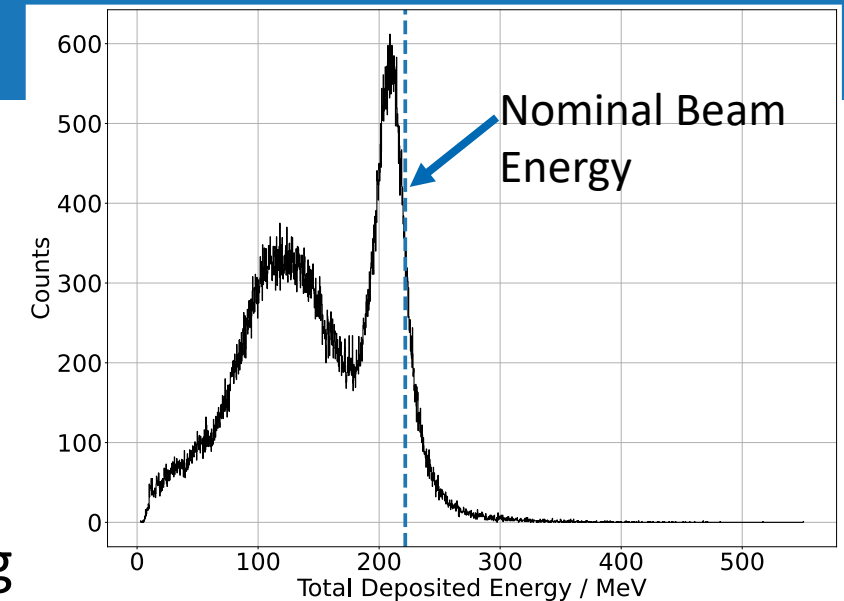
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- Protons classified as multiple energy deposition events within a 50 ns global coincidence window
 - Main peak of total energy deposition using in analysis
- Energy deposition per channel unfolded
 - Accounts for photon statistics and microcell saturation

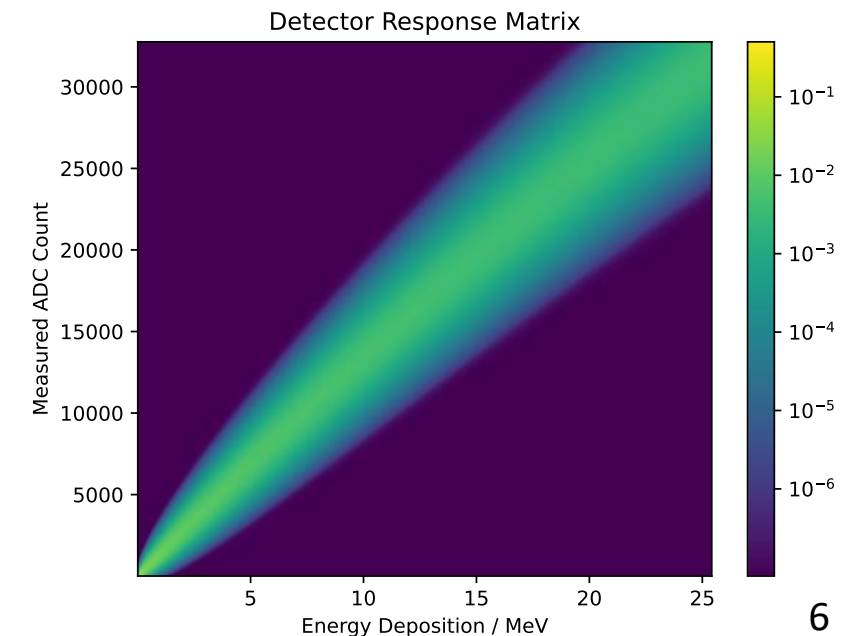
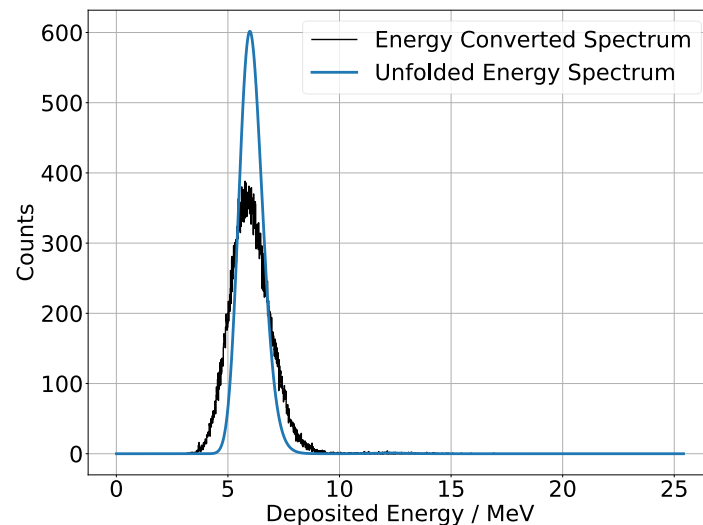


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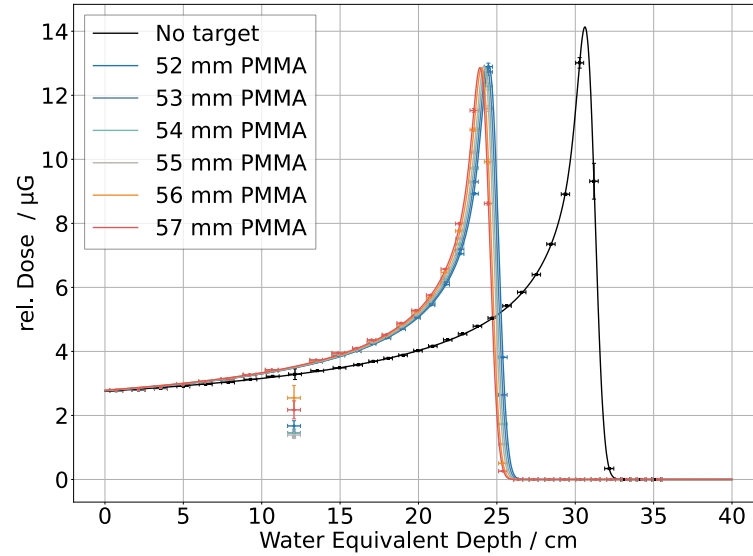
- Protons classified as multiple energy deposition events within a 50 ns global coincidence window
 - Main peak of total energy deposition using in analysis
- Energy deposition per channel unfolded
 - Accounts for photon statistics and microcell saturation
- Detector response function used for Bayesian unfolding



$$R_{ij} = P(A_i | E_j) \approx \sum_{k=0}^{k_{\max}} \binom{N_{\text{cells}}}{k} p(E_j)^k (1 - p(E_j))^{N_{\text{cells}} - k} \cdot \mathcal{N}(A_i - k \cdot \mu_{\text{SPE}}, 0, \sqrt{k} \sigma_{\text{SPE}})$$



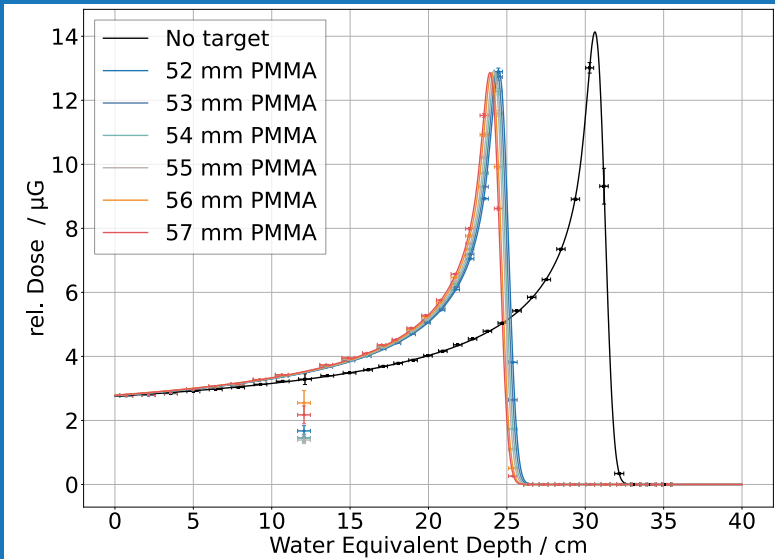
Results



- Fitted using analytical approximation of T. Bortfeld (1997)

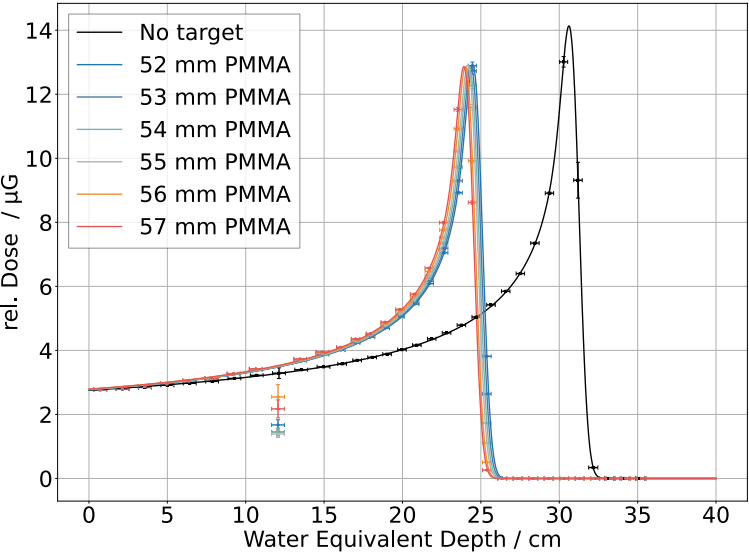
Results

- Fitted using analytical approximation of T. Bortfeld (1997)
- WET Phantom thickness reconstruction within 0.3 mm (0.5 %)



Phantom	R_0 / cm	σ / cm	t / cm	$t_{\text{nom.}}$ / cm	Δt / μm	$\Delta t_{\%}$ / %
None	30.997 ± 0.002	0.453 ± 0.005	-	-	-	-
52 mm	24.850 ± 0.003	0.449 ± 0.003	6.147 ± 0.004	6.136	110	0.18
53 mm	24.717 ± 0.004	0.445 ± 0.002	6.280 ± 0.004	6.254	260	0.41
54 mm	24.601 ± 0.004	0.441 ± 0.003	6.396 ± 0.004	6.372	240	0.38
55 mm	24.493 ± 0.004	0.435 ± 0.003	6.504 ± 0.004	6.490	140	0.22
56 mm	24.362 ± 0.004	0.441 ± 0.006	6.635 ± 0.004	6.608	270	0.41
57 mm	24.253 ± 0.003	0.440 ± 0.008	6.744 ± 0.003	6.726	180	0.27

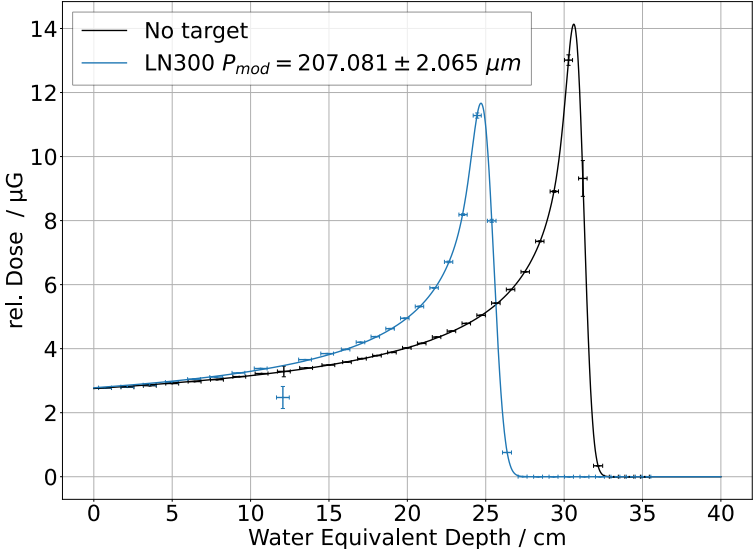
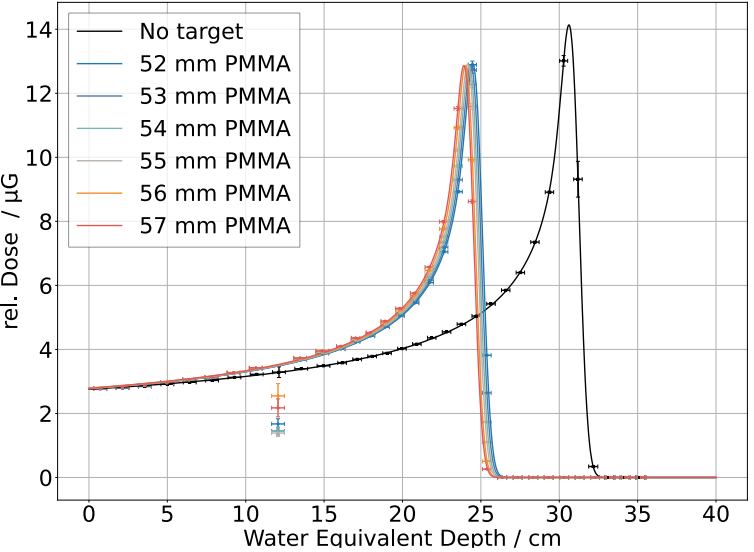
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- Fitted using analytical approximation of T. Bortfeld (1997)
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 - ± 0.4 mm to 0.5 mm for IBA Giraffe/Zebra/Lynx, Sun Nuclear DQA3

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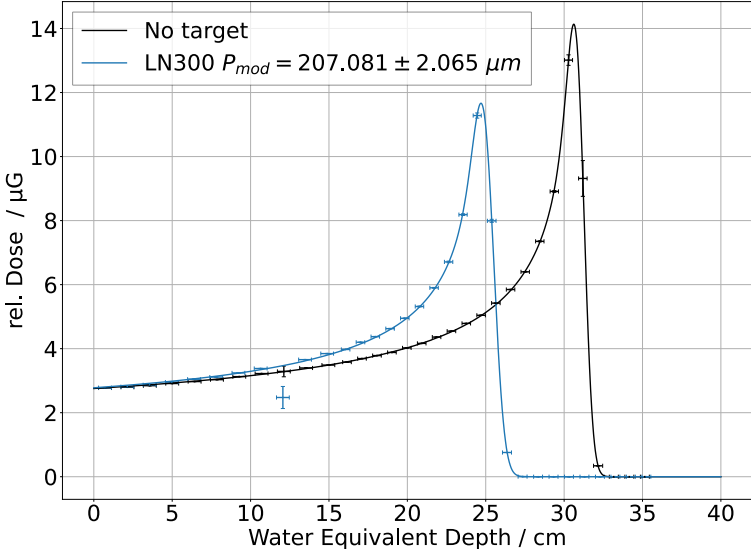
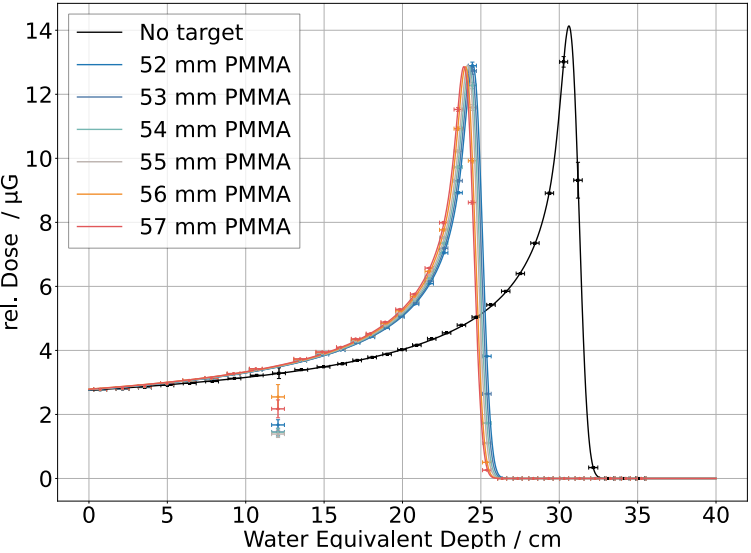
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- $P_{mod, meas.} = 207.1 \mu\text{m}$, $P_{mod, nom.} = 210 \mu\text{m}$

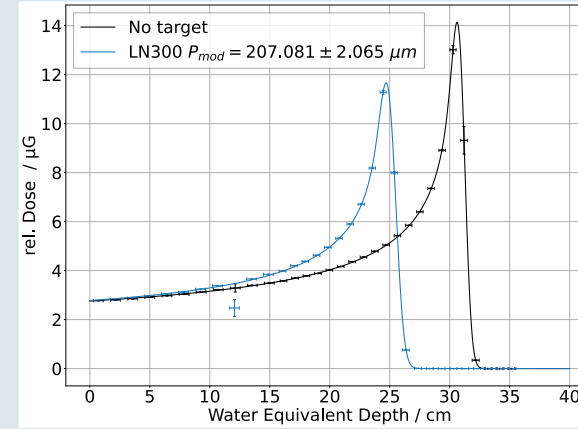
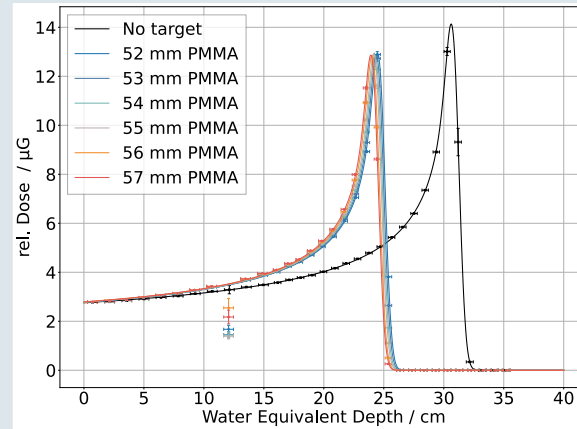
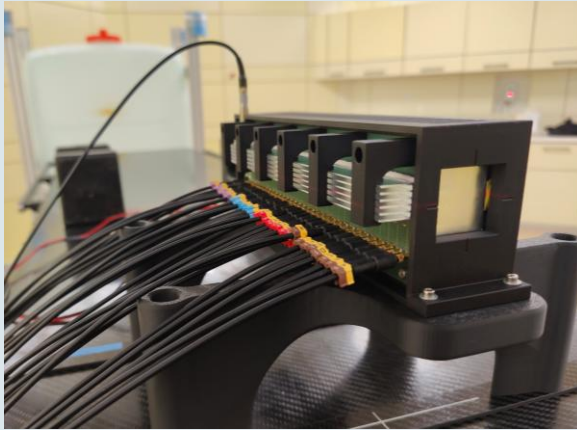
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- $P_{\text{mod, meas.}} = 207.1 \mu\text{m}$, $P_{\text{mod, nom.}} = 210 \mu\text{m}$
- Cumulative phantom energy deposition $< 2 \mu\text{J}$
 - Reducible by at least a factor 10

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Thank you for your attention!

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